Chapter 1: Cellular Biology

The cell is defined as the basic structural and functional unit of all living organisms. Every organism is composed of one or more cells. The cell theory is a fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells. The nucleus is the membrane-bound organelle that contains the genetic material. The mitochondria is the organelle responsible for producing ATP through cellular respiration.

Chapter 2: Photosynthesis

Photosynthesis is defined as the process by which green plants transform light energy into chemical energy. Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll. Chlorophyll is the pigment that absorbs light energy. The overall equation for photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen. The summary equation of photosynthesis is $6CO_2 + 6H_2O = C_6H_{12}O_6 + 6O_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH. The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

Chapter 3: Cellular Respiration

Cellular respiration is the process by which cells break down glucose to produce ATP. Cellular respiration consists of three main stages. Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate. The Krebs cycle occurs in the mitochondrial matrix. The electron transport chain is the final stage and occurs on the inner mitochondrial membrane. Fermentation is the anaerobic process that regenerates NAD^+ when oxygen is unavailable. The summary equation of cellular respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.

Chapter 4: Cell Transport

The cell membrane is the selectively permeable barrier that controls what enters and leaves the cell. Diffusion is the passive movement of molecules from an area of high concentration to low concentration. Osmosis is defined as the diffusion of water across a selectively permeable membrane. Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP. Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

Chapter 5: DNA and Protein Synthesis

DNA is defined as the molecule that carries the genetic instructions for the development and functioning of living organisms. DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules. Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA. Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence. Each three-nucleotide codon on mRNA specifies one amino acid. The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

Chapter 6: Cell Division

Mitosis is the process of nuclear division that produces two genetically identical daughter cells. The cell cycle consists of interphase, mitosis, and cytokinesis. Interphase is the stage during which the cell grows, copies its DNA, and prepares for division. Prophase is the first stage of mitosis, during which chromosomes condense. Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell. Meiosis is defined as the specialized cell division that produces four genetically diverse haploid gametes.

Chapter 7: Genetics

A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait. An allele is an alternative form of a gene at the same locus. The genotype is the genetic makeup of an organism, while the phenotype is its observable characteristics. A dominant allele is expressed when only one copy is present. A recessive allele is only expressed when two copies are present. Punnett squares are used to predict the probability of offspring inheriting particular traits.

Chapter 8: Enzymes

An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed. The active site is the region of an enzyme where the substrate binds. The induced fit model describes how the enzyme changes shape to accommodate the substrate. Optimal temperature is the temperature at which an enzyme works fastest. Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.

Chapter 9: Homeostasis

Homeostasis is defined as the maintenance of a stable internal environment despite external changes. Negative feedback is a mechanism that reverses a change, bringing the system back to its set point. Positive feedback amplifies a change and drives the system further from its set point. The normal human body temperature is maintained at approximately 37 degrees Celsius. Blood glucose concentration is regulated by the hormones insulin and glucagon. Insulin is secreted when blood glucose is high, promoting glucose uptake.

Chapter 10: Human Systems

The circulatory system transports oxygen, nutrients, and waste products throughout the body. The heart is the muscular organ that pumps blood through the circulatory system. The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide. The nervous system coordinates rapid responses through electrical signals called nerve impulses. The endocrine system regulates long-term processes through chemical messengers called hormones.